

Dataset & Research Proposal

CS577-13:Principles and Techniques of Data Science

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Section 1: Title

”Tracking the Changes in Housing Prices Over a Decade Across 50 States using Housing Sales Data to Find the Most Affordable State in this Economy.”

Section 2: Research Question

”How the housing market changed over the period of a decade across the fifty U.S. states and have these housing markets in these states remained relatively similar throughout this time frame?

Our group is aiming to investigate and describe the change over time of housing prices across the 50 states, which will include analyzing the median sale price and listing price per square foot of these houses per state in the ten-year time period. Consequently, our project can be best described as being descriptive since our project aims to describe the housing market of each state throughout the various metrics in this data set. For example one row in my dataset shows information regarding the date of the data, the state, which is California, the date in which the data was last updated by Redfin, the region type, the region id, the amount of homes sold, the median sale price, the listing price per square foot, the median days on market for a house, the amount of new listings, the amount of active listings, and the amount of pending sales.

Section 3: Dataset Information

1. Dataset source (URL)

<https://www.redfin.com/news/data-center/downloads/>

2. How the data was collected (if known)

According to the Redfin website, this data offers a detailed snapshot of U.S. residential real estate activity, which includes the full cycle of a home sale; from initial listing through closing. This data is updated on a weekly and monthly basis.

They mention that the dataset covers rolling four-week windows. They stated that aggregating over 4 weeks provides more stable values, particularly for smaller markets. They

measure year-over-year changes in the dataset of sales prices and listings. Year over year comparisons look back 52 weeks to "preserve alignment".

Lastly, they mention that all metrics cover residential properties, which includes single family homes, condo/co-op, town-house, and multi-family houses. They also note that their data is defined consistently with the US Office of Management and Budget metropolitan divisions where they exist, and metropolitan statistical areas elsewhere.

3. License or access type (public, API, Kaggle, etc.)

This data is publicly available for download.

4. Number of rows

There are over 6000 rows, which is more than sufficient to perform detailed data analysis using pandas.

5. Number of columns

There are a total of 20 columns. A lot of these columns include the year-to-year changes of other columns. For example, Median Price has a YoY Median price. This is true for many of the columns.

6. Data types (numeric, categorical, text, date)

Since there are so many columns, it is to be expected that there are a lot of data types. Without checking the type of the data using pandas, the data appears to be a mix of numerical values, date values, and string objects (categorical values like state and state names).

7. Time range (if applicable)

The time range for this dataset was something that was specified in the website. Our group chose the time frame from Jan. 2016 to Aug. 2026, which is a gap of over 10 years.

8. Known limitations of the dataset

The most obvious limitation of this dataset is that it looks at the entire state for calculating the median price of homes. Because of this, the value that is shown in the median price column of the dataset misrepresents the very expensive homes in certain regions of the state, particularly urban cities with a large population. For example, one of the rows I briefly

looked at showed that the median house pricing for California is around 700K. However, this median price does not represent affluent areas of California, such as the City of Los Angeles or San Francisco. In fact, the median pricing of houses in these areas is around one million, which is much larger than the median pricing of California shown in the dataset. Because of the aforementioned information, one looking to settle in and buy housing in urban areas of states should not solely rely on this data. Because it is likely that the data is misrepresenting prices in urban areas.

Another limitation of this dataset is that, despite the fact that it lists US territories like Guam or Puerto Rico as part of the US states, the data for the Housing Prices for these territories is unavailable. I am quite confounded at the fact these territories are present in the dataset despite the fact that I only included the 50 states when creating this dataset through the Redfin website and these territories contain no housing data so they serve no purpose in the dataset.

Section 4: Planned EDA Questions

Section 5: Anticipated Challenges